Travel Behaviour and Society 29 (2022) 319–329

Contents lists available at ScienceDirect

Travel Behaviour and Society

journal homepage: www.elsevier.com/locate/tbs

Modelling lane changing behaviors for bus exiting at bus bay stops considering driving styles: A game theoretical approach

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ARTICLE INFO

Keywords:
Public transportation
Lane changing behaviors
Driving styles
Incomplete information game
Bus bay stops
Drivers' decision makings

ABSTRACT

Bus is one of the most important travel modes for citizens, but lane changing behaviors for bus exiting at bus bay stops have negative impacts on traffic flow operations. It is necessary to investigate the characteristics of drivers' behavioral decision-makings in lane changing for bus exiting for improving public transportation operations. The interactions between passenger car drivers in target lanes and bus drivers with different driving styles (e.g., aggressive or conservative style) are analyzed based on the game theory. A two-player, non-zero-sum, non-cooperative game model with incomplete information is formulated to examine the impacts of driving styles on the lane changing behavioral decision makings. The bi-level programming approach is applied to calibrate the parameters in the formulated model with solving the Bayesian equilibria and minimizing the prediction errors. The vehicle trajectory data extracted from Lvboqiao bus bay stop, Dalian, China are used for model calibration and validation. The results reveal that the formulated model has 73.01%, 86.60% and 83.54% accuracy rates on average for predicting the passenger car, aggressive and conservative bus drivers' behaviors, respectively. At a 95% confidence level, the time error of a lane changing behavior is statistically significant by the paired *t*-test. Compared with the previous research, it can be proven that the consideration of driving styles is helpful to deeply explore the interactions between passenger car and bus drivers' behaviors. The proposed model provides decent theoretical support for the drivers to make better decision-makings, and improves the traffic safety and efficiency around bus bay stops.

1. Introduction

Bus operations are being paid more and more attentions by governments in several decades in order to increase travel efficiency and decrease carbon emissions. Bus bay stops are a kind of initiatives to improve the efficiency of public transportation, especially when dedicated bus lanes cannot be implemented due to limited road resources. Compared with lane changing for bus entering, when buses exit from bus bay stops, all the bus drivers have to change lanes compulsively at almost fixed points (i.e., bus bay stops). In this case, the bus drivers' lane changing behaviors for bus exiting bring more difficulties for bus drivers to make decisions, and cause more negative effects on the traveling of passenger cars on the adjacent lanes. Therefore, it is of great significance to explore the mandatory lane changing behaviors for bus exiting at bus bay stops to improve the efficiency and safety of public transportation.

Lane changers are regarded as the main research objects in the rule-based model (e.g., Gipps, 1986) and the utility theory-based model (e.g., Zhou et al., 2019), but in fact passenger cars on the adjacent lanes near bus bay stops cannot be neglected during the process of bus lane changing. It is because that the drivers' behaviors for passenger cars on the adjacent lanes can affect the bus drivers' decision makings, and the bus drivers have to respond to the passenger car drivers' behaviors to guarantee the safety of driving. Hence, the passenger car and bus drivers' behaviors should be both taken into consideration.

Another issue of lane changing models is that drivers' driving styles are often neglected. Drivers who have different driving styles may make different decisions when meeting the same situations (Tehran et al., 2016). In addition, the same driver may have different driving styles when he/she meets different temporal and spatial traffic situations (Li et al., 2019). Drivers' driving styles are not always constant but sometimes random. Through adding such effects into the lane changing modeling, the predicted behaviors of both passenger car and bus drivers

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https://doi.org/10.1016/j.tbs.2022.07.008
Received 4 April 2021; Received in revised form 2 July 2022; Accepted 15 July 2022
Available online 28 July 2022
2214-367X/

will reflect the actual behaviors more accurately.

The objective of this paper is to formulate a new model of mandatory lane changing for bus exiting in the context of bus bay stops by considering driving styles. Firstly, a game theory-based approach is utilized to explore the behaviors of the lane changer (i.e., bus driver) and the driver of the passing vehicle near the bus bay stop (i.e., passenger car driver), and the interactions between them. More specifically, the Harsanyi theory is employed to investigate the effects of driving styles on drivers' behaviors and decision makings in an incomplete information game. Then, the formulated model is calibrated using the bi-level programming approach, by which the lower level is the Bayesian equilibrium and the upper level aims at minimizing the deviation between the observed and predicted behaviors. Finally, a case study is accomplished to verify the performance of the formulated model.

The remainder of this paper is organized as below. Section 2 presents the literature review about the existing lane changing models and their limitations. Section 3 explains the methodology. The calibration and validation of the formulated model are introduced in Section 4. The case study is elaborated in Section 5. Section 6 concludes the main contributions of this paper and proposes the recommendations for the future research.

2. Literature review

2.1. Rule-based lane changing models

Gipps (1986) firstly proposed the lane changing decision model, in which obstructions, speed advantages, heavy vehicles etc. deterministic conditions were considered. Based on Gipps's model, scholars made some improvements with adding other factors in the model, such as the probability of changing a lane (Yang and Koutsopoulos, 1996), the maximum acceptable decrease of speed for the follow vehicle and the time for the lane changer before reaching the end of the lane (Hidas, 2002, 2005), and the changes of vehicle accelerations (Kesting et al., 2007).

In the rule-based models, a series of judgement principles of lane changing decision making were developed, including temporal (e.g., speed and density) and spatial (e.g., acceptance gap and roadblock) characteristics of traffic conditions. However, drivers' driving styles which are ignored in the literatures discussed above play a vital role in the lane changing decision making since the judgement of the lane changing is subjective to some extent.

2.2. Lane changing models considering driving styles

For the lane changing behaviors, the driving styles were commonly divided into two categories, that is, aggressive and conservative driving styles (Sun and Elefteriadou, 2012; Zhou et al., 2019; Pang et al., 2020). Huo et al. (2020) proposed that the aggressive driving style was characterized by short time of decision making and high efficiency, but high risk of safety in lane changing behaviors. Whereas, the driver with a conservative driving style was more inclined to ensure the safety of lane changing, even if more time might be consumed.

Sun and Elefteriadou (2011, 2012) determined the driver's aggressiveness by applying a focus group and the evaluations of himself/herself and his/her friends. The K-means method was also applied to classify the driving styles through some deterministic rules set in advance (Ren et al., 2019; Li et al., 2020). As stated earlier, each driver's choice of lane changing has randomness at the moment that the driver chooses a strategy, because even the same driver, he/she may make different strategic choices in different situations. To reveal the randomness in lane changing decision making, scholars applied the random utility theory and employed different variables to reflect the driving styles, including speed difference (Zhou et al., 2019), the characteristics of speed and space (Pang et al., 2020), and gap acceptance (Sun and Elefteriadou, 2014).

In the existing literatures of lane changing models considering driving styles, scholars developed some deterministic rules to predict drivers' driving styles, and regarded the lane changer as the main research object. But actually, the vehicles in the target lanes will affect and also be affected by the lane changer. Therefore, it is necessary to analyze the interactions between the lane changer and the vehicle on the target lanes.

2.3. Game theory-based lane changing models

Researchers have utilized different game theory-based approaches to study the issue of lane changing on highways or urban streets. Kita (1999) first defined the lane changing behaviors in the context of highway as a two-player mixed strategy game with complete information, and considered the time to collision in the payoff functions. Based on Kita's research, scholars added other decision factors, like minimum travel time, accepted acceleration (Kang and Rakha, 2017), and conflict risks (Arbis and Dixit, 2019) in the game model on highways. As for the lane changing behaviors on urban streets, Shao et al. (2018) focused on the illegal lane changing behaviors for electric bicycles. Besides the mixed strategy game, other game models were developed to enrich the understanding of lane changing behaviors, such as repeated game (Kang and Rakha, 2018), cooperative game (Lin et al., 2019; Wang et al., 2015), and Stackelberg game (Yu et al., 2018).

When some game information is not accurately known by other players, an incomplete information game will be formed. To capture the uncertainty in the incomplete information game, the Harsanyi theory is usually utilized to transform a game of incomplete information into a game of imperfect information (Harsanyi, 1967). The Harsanyi theory was used to randomly choose the types of lane changing (Talebpour et al., 2015), driving environment (Ali et al., 2019) and driving styles (Shao et al., 2020) in the context of lane changing on highways.

The game theory-based existing researches on lane changing mostly focus on the context of highways. The context of urban streets is rarely focused. The lane changing behaviors for bus exiting at bus bay stops, as an essential part on urban streets, need to be paid attention to. It is necessary to consider different driving styles in lane changing issues, since drivers' driving styles have great influences on their subjective behavioral decisions. Therefore, the mandatory lane changing behaviors for bus exiting at bus bay stops from the perspective of driving styles are studied in this paper.

3. Methodology

This paper aims at formulating a model to describe the lane changing behaviors of bus drivers with different driving styles for bus exiting at bus bays stops without dedicated bus lanes. Under this situation, neither the buses nor the passenger cars have traffic priorities, thus the players' behaviors cannot be predicted by each other in advance.

Passenger car and bus drivers both have different driving styles, but their driving styles have varying degrees of influences on their actions in the process of lane changing. The passenger car drivers with aggressive driving styles are willing to take risks in the lane changing process to pursue efficiency, and they are inclined to the efficient strategy, i.e., the strategy of not giving way. Because of the greater economic loss caused by collision conflicts, the aggressive passenger car driver will not directly compete with the bus driver when the strategy of changing a lane is chosen by the bus driver. In this case, the passenger car driver, as a rational player, is less aggressive in the game. The driving style of the passenger car driver will not be revealed obviously in the process of bus lane changing, and accordingly has limited effects on behavioral decision makings. Thus, the bus drivers' driving styles are only considered in this study.

3.1. Game and its components

The mandatory lane changing model for the bus driver with an aggressive driving style (B_AS hereon) is firstly developed, and then that for the bus driver with a conservative driving style (B_CS hereon) is formulated (Huo et al, 2020). The lane changing situations under different driving styles are illustrated in Fig. 1. As shown in Fig. 1(a), when the follow vehicle has arrived at or passed through point A, which means that the follow vehicle has entered the bus bay stop; at this time, B_AS is more likely to change a lane to become the new leader even if traffic situations are not satisfying. As presented in Fig. 1(b), B_CS may not be inclined to change a lane due to the limited acceptance gap; at this time, he/she will wait the next opportunity for changing a lane after the follow vehicle has passed through the bus bay stop completely.

The lane changing behavior can be modeled as a two-player, non-zero-sum, non-cooperative game with incomplete information. The passenger car that is closest to the bus bay stop is the first concern because the interactions between such a passenger car and the bus are dominant (Kita, 1999). Therefore, the passenger car and bus drivers are the two players in the game. Each player will receive a payoff according to his/her action, and the sum of the two players' payoffs is not zero. In a non-cooperative game with incomplete information, players have inaccurate information and have to make predictions about other players' actions. Since a passenger car driver does not know the driving style of the bus driver beforehand, the bus driver's driving style is not the common knowledge of the two players, thus the game is an incomplete information game.

In the game between the passenger car and bus drivers, the passenger car driver (S) has two strategies, that is, giving way to the bus (S_1) and not giving way to the bus (S_2). No matter the aggressive or conservative bus driver (B), he/she has two strategies, that is, changing a lane (B_1) and not changing a lane (B_2). The strategic form game between the passenger car and bus drivers with the concern of bus drivers' different driving styles is shown in Table 1. $R_{ij},\ i,j \in \{1,2\}$ denotes the payoff of the strategy which the passenger car driver chooses in different strategy profiles; $P_{ij},\ i,j \in \{1,2\}$ means the payoff of the strategy which B_AS chooses in different strategy profiles; and $Q_{ij},\ i,j \in \{1,2\}$ represents the payoff of the strategy which B_CS chooses in different strategy profiles.

In this study, the Harsanyi transformation (Harsanyi, 1967) is utilized to transform a game of incomplete information into a game of imperfect information. This approach is to introduce "Nature" as a player who chooses the type of each bus driver's driving style. The "Nature" can be regarded as another player in the game without payoffs. The incomplete information game between the passenger car drivers and the bus drivers with different driving styles is therefore transformed into an imperfect information game between the passenger car drivers and the Nature's actions. The extensive game with imperfect information after the transformation is illustrated in Fig. 2. In this figure, based on the extensive game with incomplete information, the player "Nature" is added at the root node. The "Nature" chooses the type of the bus driver's driving style randomly. Assume that it chooses the aggressive driving style with probability p and the conservative style with

Table 1
Strategic form game between passenger car and bus drivers.

S	B			
	B_AS		B_CS	
	B_1	B_2	B_1	B_2
S_1	$R_{11},\ P_{11}$	$R_{12},\ P_{12}$	$R_{11},\ Q_{11}$	$R_{12},\ Q_{12}$
S_2	$R_{21},\ P_{21}$	$R_{22},\ P_{22}$	$R_{21},\ Q_{21}$	$R_{22},\ Q_{22}$

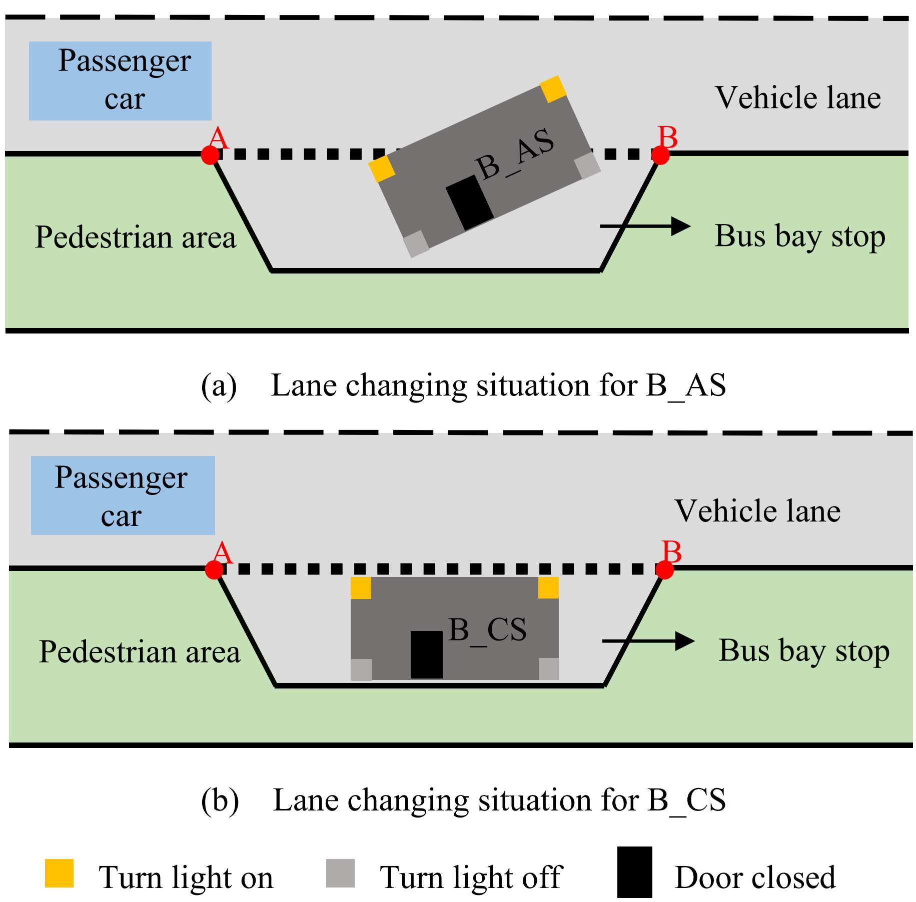


Fig. 1. Lane changing situations for bus drivers with different driving styles.

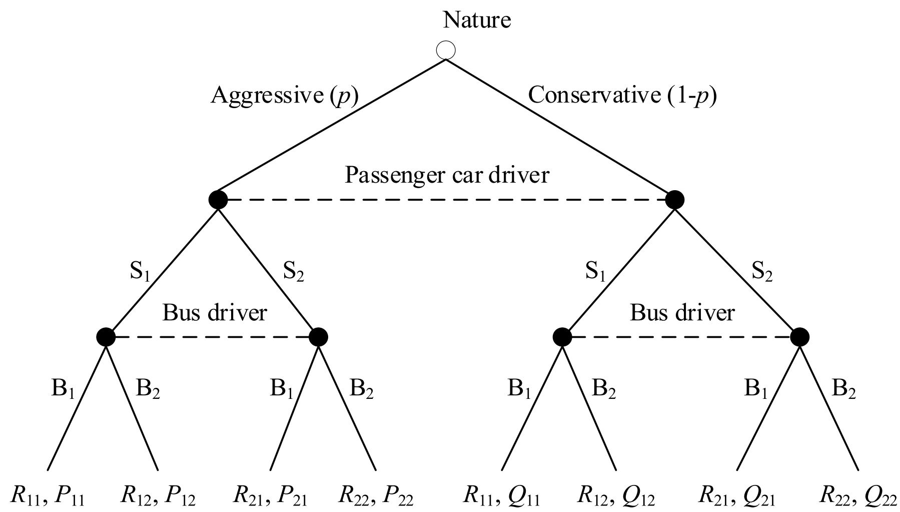


Fig. 2. Extensive game with imperfect information.

probability $1 - p$. The driving styles are not the common knowledge for the passenger car and bus drivers but a kind of private information for the bus driver himself/herself. The passenger car driver does not have an idea about the actual state (i.e., the bus driver's driving style). The bus driver knows the actual state, that is, he/she is clear about his/her own driving style. In each state, the passenger car and bus drivers both have two strategies. It should be noted that all the players involved in the game have the consistent judgements about the probability of the Nature's actions. Through utilizing the Harsanyi transformation, the transformed game between the passenger car and bus drivers can be obtained, as presented in Table 2. $B_k^A,\ k \in \{1,2\}$ is the strategy for B_AS to choose, and $B_k^C,\ k \in \{1,2\}$ is the strategy for B_CS to choose. In the two-player game discussed here, both the passenger car and bus drivers have two strategies, thus there are totally eight sets of strategy profiles.

In Table 2, $O_{ij},\ \ i \in \{1,2\}, j \in \{1,2,3,4\}$ is the expected payoff of the strategy which the passenger car driver chooses in different strategy profiles with incomplete information, and it can be calculated as.

$$\begin{cases} O_{i1} = R_{i1} \\ O_{i2} = p \cdot R_{i1} + (1-p) \cdot R_{i2} \\ O_{i3} = (1-p) \cdot R_{i1} + p \cdot R_{i2} \\ O_{i4} = R_{i2} \end{cases},\quad i \in \{1,2\} \tag{1}$$

3.2. Payoff functions

The payoffs for the passenger car and bus drivers are established from the perspective of efficiency and safety. Describing the payoff functions with more variables makes them more comprehensive. However, when there are too many variables, the calibration of the parameters is not computationally practical (Talebpour et al, 2015). The speed, acceleration and time of the vehicles are the variables commonly applied in the payoff functions (Ali et al, 2019; Arbis and Dixit, 2019; Kita, 1999). Therefore, the payoffs that are related to efficiency are constructed by considering the acceleration and speed of the passenger car and bus, while the payoffs that are related to safety are built using the time that the vehicles consume at bus bay stops. It should be noted that the traffic delay for the mass passengers carried by the bus will raise, especially when the bus driver chooses the strategy of not changing a lane.

Table 2
Transformed game between passenger car and bus drivers.

S	B			
	$B_1^A B_1^C$	$B_1^A B_2^C$	$B_2^A B_1^C$	$B_2^A B_2^C$
S_1	O_{11}, (P_{11}, Q_{11})	O_{12}, (P_{11}, Q_{12})	O_{13}, (P_{12}, Q_{11})	O_{14}, (P_{12}, Q_{12})
S_2	O_{21}, (P_{21}, Q_{21})	O_{22}, (P_{21}, Q_{22})	O_{23}, (P_{22}, Q_{21})	O_{24}, (P_{22}, Q_{22})

3.2.1. Payoffs for passenger car drivers

For the strategy of giving way to the bus, the passenger car driver shows courtesy but loses time. The time that the bus driver needs to finish exiting from the bus bay stop is taken into account in the payoff function. For the strategy of not giving way to the bus, the acceleration of the passenger car when it arrives at the bus bay stop is considered in the payoff function. In the strategy profile of not giving way and changing a lane, the acceleration of the passenger car when it leaves the bus bay stop is also used to represent the collision risk among the nearby vehicles. In the complete information game, the payoffs of the passenger car driver for different strategy profiles can be expressed as.

$$\begin{cases} R_{11} = \alpha_{11,0} + \alpha_{11,1} T_B \\ R_{12} = \alpha_{12,0} + \alpha_{12,1} T_B \\ R_{21} = \alpha_{21,0} + \alpha_{21,1} E_{SA} + \alpha_{21,2} E_{SL} \\ R_{22} = \alpha_{22,0} + \alpha_{22,1} E_{SA} \end{cases} \tag{2}$$

where T_B (s) is the time that the bus driver consumes to exit from the bus bay stop; E_{SA} (m/s^2) is the acceleration of the passenger car when it arrives at the bus bay stop; E_{SL} (m/s^2) is the acceleration of the passenger car when it leaves the bus bay stop; $\alpha = \{\alpha_{11,0}, \alpha_{11,1}, \alpha_{12,0}, \alpha_{12,1}, \alpha_{21,0}, \alpha_{21,1}, \alpha_{21,2}, \alpha_{22,0}, \alpha_{22,1}\}$ is a set of undetermined parameters.

3.2.2. Payoffs for B_AS

B_AS pays more attention to the efficiency of lane changing for exiting. When the passenger car arrives at the bus bay stop, if the bus has already started (i.e., the speed of the bus is more than zero), B_AS will immediately insert to the gap between the two passenger cars and change to the adjacent lane. The speed of the bus is taken into consideration in the payoff of the strategy of changing a lane. When the arriving acceleration of the passenger car is relatively large, it is easy for the bus driver to perceive the obvious change of the speed of the passenger car. At this time, the bus driver, as a rational decision maker, will not change a lane to avoid collisions. Therefore, the arriving acceleration of the passenger car is considered in the payoff of the strategy of not changing a lane. In the strategy profile of not giving way and changing a lane, the acceleration of the passenger car when it arrives at the bus bay stop is also applied to represent the collision risk between the passenger car and bus. The payoffs for B_AS for different strategy profiles can be given by.

$$\begin{cases} P_{11} = \beta_{11,0} + \beta_{11,1} V_{\mathrm{BL}} \\ P_{12} = \beta_{12,0} + \beta_{12,1} E_{\mathrm{SA}} \\ P_{21} = \beta_{21,0} + \beta_{21,1} V_{\mathrm{BL}} + \beta_{21,2} E_{\mathrm{SA}} \\ P_{22} = \beta_{22,0} + \beta_{22,1} E_{\mathrm{SA}} \end{cases} \tag{3}$$

where V_{BL} (m/s) is the speed of the bus when the passenger car arrives at the bus bay stop; $\beta = \{\beta_{11,0}, \beta_{11,1}, \beta_{12,0}, \beta_{12,1}, \beta_{21,0}, \beta_{21,1}, \beta_{21,2}, \beta_{22,0}, \beta_{22,1}\}$ is a set of undetermined parameters.

3.2.3. Payoffs for B_CS

Compared with B_AS, B_CS focuses more on the collision risk of changing a lane, and the conservative driving style is friendly for controlling the starting acceleration and speed. If the B_CS decides to change a lane when the passenger car arrives at the bus bay stop, the traffic condition should be very likely to be favorable for the bus driver. The speed of the bus is used in the payoff of the strategy of changing a lane. When the bus driver decides to wait at the bus bay stop, he/she will change a lane after the current passenger car passes through the bus bay stop completely. Therefore, the time that the passenger car driver consumes to pass through the bus bay stop is utilized to calculate the payoff of the strategy of not changing a lane. The number of passengers carried by the bus is also considered in the payoff. In the strategy profile of not giving way and changing a lane, the acceleration of the passenger car when it arrives at the bus bay stop is also applied to represent the collision risk. The payoffs for B_CS for different strategy profiles can be obtained by.

$$\begin{cases} Q_{11} = \delta_{11,0} + \delta_{11,1} V_{\mathrm{BL}} \\ Q_{12} = \delta_{12,0} + \delta_{12,1} T_{\mathrm{S}} \times L \\ Q_{21} = \delta_{21,0} + \delta_{21,1} V_{\mathrm{BL}} + \delta_{21,2} E_{\mathrm{SA}} \\ Q_{22} = \delta_{22,0} + \delta_{22,1} T_{\mathrm{S}} \times L \end{cases} \tag{4}$$

where T_{S} (s) is the time that the passenger car driver consumes to pass through the bus bay stop; L (person) is the number of passengers carried by the bus; $\delta = \{\delta_{11,0}, \delta_{11,1}, \delta_{12,0}, \delta_{12,1}, \delta_{21,0}, \delta_{21,1}, \delta_{21,2}, \delta_{22,0}, \delta_{22,1}\}$ is a set of undetermined parameters.

3.3. Equilibrium analysis

All the players in the game pursue the maximum payoffs. When the players' strategies are the best responses to each other, the Bayesian equilibrium, a solution point where no player can increase his/her payoff by changing his/her strategic choice, can be generated. The "Nature" chooses a specific type of the bus driver (c_{B}) from a type space of the bus driver C_{B}, i.e.,$c_{\mathrm{B}} \in C_{\mathrm{B}}$. Every player in the game chooses his/her own action which is denoted by a_{S} and a_{B} for the passenger car and bus drivers, from the action sets A_{S} and A_{B}, respectively, i.e., $a_{\mathrm{S}} \in A_{\mathrm{S}}$ and $a_{\mathrm{B}} \in A_{\mathrm{B}}$. When the strategies for the bus driver with different driving styles (s_{B}) are given, the expected payoff for the passenger car driver choosing a_{S} is expressed as $v_{\mathrm{S}}(a_{\mathrm{S}}, s_{\mathrm{B}})$. The Bayesian equilibrium for the passenger car and bus drivers, denoted by a_{S}^* and a_{B}^*, can be given by.

$$\begin{cases} a_{\mathrm{S}}^* \in \underset{a_{\mathrm{S}} \in A_{\mathrm{S}}}{\operatorname{argmax}} v_{\mathrm{S}}(a_{\mathrm{S}}, s_{\mathrm{B}}) \\ v_{\mathrm{S}}(a_{\mathrm{S}}, s_{\mathrm{B}}) = \sum_{c_{\mathrm{B}} \in C_{\mathrm{B}}} p(c_{\mathrm{B}}) \cdot u_{\mathrm{S}}[a_{\mathrm{S}}, a_{\mathrm{B}}^*(c_{\mathrm{B}})] \end{cases} \tag{5}$$

$$a_{\mathrm{B}}^*(c_{\mathrm{B}}) \in \underset{a_{\mathrm{B}} \in A_{\mathrm{B}}(c_{\mathrm{B}})}{\operatorname{argmax}} u_{\mathrm{B}}(a_{\mathrm{B}}, a_{\mathrm{S}}^*; c_{\mathrm{B}}) \tag{6}$$

where u_{S} and u_{B} represent the expected payoffs for the passenger car and bus drivers at the Bayesian equilibrium, respectively; $p(c_{\mathrm{B}})$ is the probability of the "Nature" choosing the bus driver's specific driving style; argmax is a function which aims at solving the optimal values of a_{S}^* and a_{B}^*.

The players in the game choose their strategies with uncertainty. The probability of choosing the strategy of changing a lane for B_AS is denoted by x, that is, $P(\mathrm{B}_1|\mathrm{B\_AS}) = x$. It is indicated that there is still a certain risk in the strategy of changing a lane. That is to say, the ideal conditions for changing lanes are not met, but the aggressive driving style makes the driver willing to take this risk. In this case, for the conservative driver who pays more attention to safety, the driver will definitely not choose the strategy of changing a lane (Fudenberg and Tirole, 2010). The probability of the strategy of giving way for the passenger car driver is denoted by y, that is, $P(\mathrm{S}_1) = y$. There are totally 4 scenarios, that is, B_AS chooses the strategy of changing a lane, B_AS chooses the strategy of not changing a lane, B_CS chooses the strategy of changing a lane and B_CS chooses the strategy of not changing a lane. Based on the probability of the "Nature" choosing B_AS (PN hereon) and the Bayesian theorem, the probabilities of the strategies of changing a lane and not changing a lane for the bus driver with different driving styles can be derived as.

$$\begin{cases} P(\mathrm{B}_1, \mathrm{B\_AS}) = p \cdot x \\ P(\mathrm{B}_2, \mathrm{B\_AS}) = p \cdot (1 - x) \\ P(\mathrm{B}_1, \mathrm{B\_CS}) = 0 \\ P(\mathrm{B}_2, \mathrm{B\_CS}) = (1 - p) \end{cases} \tag{7}$$

The payoffs of the strategies of giving way and not giving way for the passenger car driver, denoted by u_{S_1} and u_{S_2}, can be given by.

$$u_{\mathrm{S}_1} = (R_{11} - R_{12}) \cdot p \cdot x + R_{12} \tag{8}$$

$$u_{\mathrm{S}_2} = (R_{21} - R_{22}) \cdot p \cdot x + R_{22} \tag{9}$$

When $u_{\mathrm{S}_1} = u_{\mathrm{S}_2}$, the equilibrium probability of the strategy of changing a lane for B_AS, denoted by x^*, can be determined by.

$$x^* = \frac{R_{22} - R_{12}}{p \cdot (R_{11} + R_{22} - R_{12} - R_{21})} \tag{10}$$

When the actual probability of the strategy of changing a lane for B_AS is more than the equilibrium probability, the payoff of the strategy of giving way for the passenger car driver is greater; when the actual probability is less than the equilibrium probability, the payoff of the strategy of not giving way for the passenger car driver is greater; when the actual probability is equal to the equilibrium probability, the passenger car driver will choose his/her strategy randomly without difference.

Accordingly, the equilibrium probability of the strategy of giving way for the passenger car driver can be determined by the payoff of B_AS, that is.

$$u_{\mathrm{B}_1} = (P_{11} - P_{21}) \cdot y + P_{21} \tag{11}$$

$$u_{\mathrm{B}_2} = (P_{12} - P_{22}) \cdot y + P_{22} \tag{12}$$

where u_{B_1} and u_{B_2} are the payoffs of the strategies of changing a lane and not changing a lane for B_AS, respectively.

When $u_{\mathrm{B}_1} = u_{\mathrm{B}_2}$, the equilibrium probability of the strategy of giving way for the passenger car driver, denoted by y^*, can be solved by.

$$y^* = \frac{P_{22} - P_{21}}{P_{11} + P_{22} - P_{12} - P_{21}} \tag{13}$$

When the actual probability of giving way to the bus is larger than the equilibrium probability, the payoff of changing a lane for B_AS is greater; when the actual probability of giving way to the bus is less than the equilibrium probability, the payoff of not changing a lane for B_AS is greater; when the actual probability equals the equilibrium probability, there is no difference between the strategies of changing a lane and not changing a lane.

4. Model calibration and validation

4.1. Model calibration

The calibration of the parameters in a game model is a challenging task because of the complexity of the game. The objective of the calibration is to minimize the difference between the observed and predicted behaviors. The framework of parameter calibration (Kang and Rakha, 2017; Ali et al., 2019) is illustrated in Fig. 3. The calibration of the parameters is translated into a bi-level programing problem, which consists of the upper and lower levels.

The upper level programing aims at minimizing the squared difference between the observed and predicted behaviors, as shown in Eq. (14). The optimization function 'optim' in R 3.6.3 is applied to optimize the objective function in Eq. (14).

$$\text{minimize} \sum_{n=1}^{N}\left[(M_n - \widehat{M}_n)^2 + (N_n - \widehat{N}_n)^2\right] \quad (14)$$

where M_n and N_n are the observed behaviors of the passenger car and bus drivers for the n th set of data in the calibration dataset; $\widehat{M}_n$ and $\widehat{N}_n$ are the predicted behaviors of the passenger car and bus drivers for the n th set of data in the calibration dataset; N is the number of samples in the calibration dataset.

The lower-level programming aims at solving the Bayesian equilibrium. Some scholars used the methods like Gambit (McKelvey et al., 2014) and Nashpy package (Nisan et al., 2007) to solve a two-tuple Nash equilibrium. However, the Bayesian equilibrium discussed here is a triad, i.e.,$[\mathrm{S}_i, (\mathrm{B}_i^{\mathrm{A}}, \mathrm{B}_i^{\mathrm{C}})]$, $i \in \{1, 2\}$, which is more complex than a two-tuple. With the logic of underlining the maximum payoff in the players' strategy sets, when the strategies of the passenger car driver, B_AS and B_CS are the best responses to other players, the Bayesian equilibrium can be obtained. Whether each strategy profile is a Bayesian equilibrium requires a complete judgement from the beginning to the end, since multiple Bayesian equilibria may exist. It should be noted that: when PN is predetermined, the Bayesian equilibria will be different with the variations of the players' payoffs resulted by the other variables; when PNs are different, the Bayesian equilibria will also vary and should be analyzed separately.

4.2. Model validation

To validate the prediction accuracy of the mandatory lane changing model formulated in this paper, the mean absolute error (MAE) is used. The calculation of MAE is given in Eq. (15). If the observed and predicted behaviors are the same, $U_m - \widehat{U}_m$ is equal to zero. Otherwise, $U_m - \widehat{U}_m$ is equal to 1.

$$\mathrm{MAE} = \frac{1}{M}\sum_{m=1}^{M}(U_m - \widehat{U}_m) \quad (15)$$

where U_m is the observed behavior for the m th set of data in the validation dataset; $\widehat{U}_m$ is the predicted behavior for the m th set of data in the validation dataset; M is the number of samples in the validation dataset.

In order to further verify the performance of the formulated model, the predicted accuracy rate of each strategy for different players should

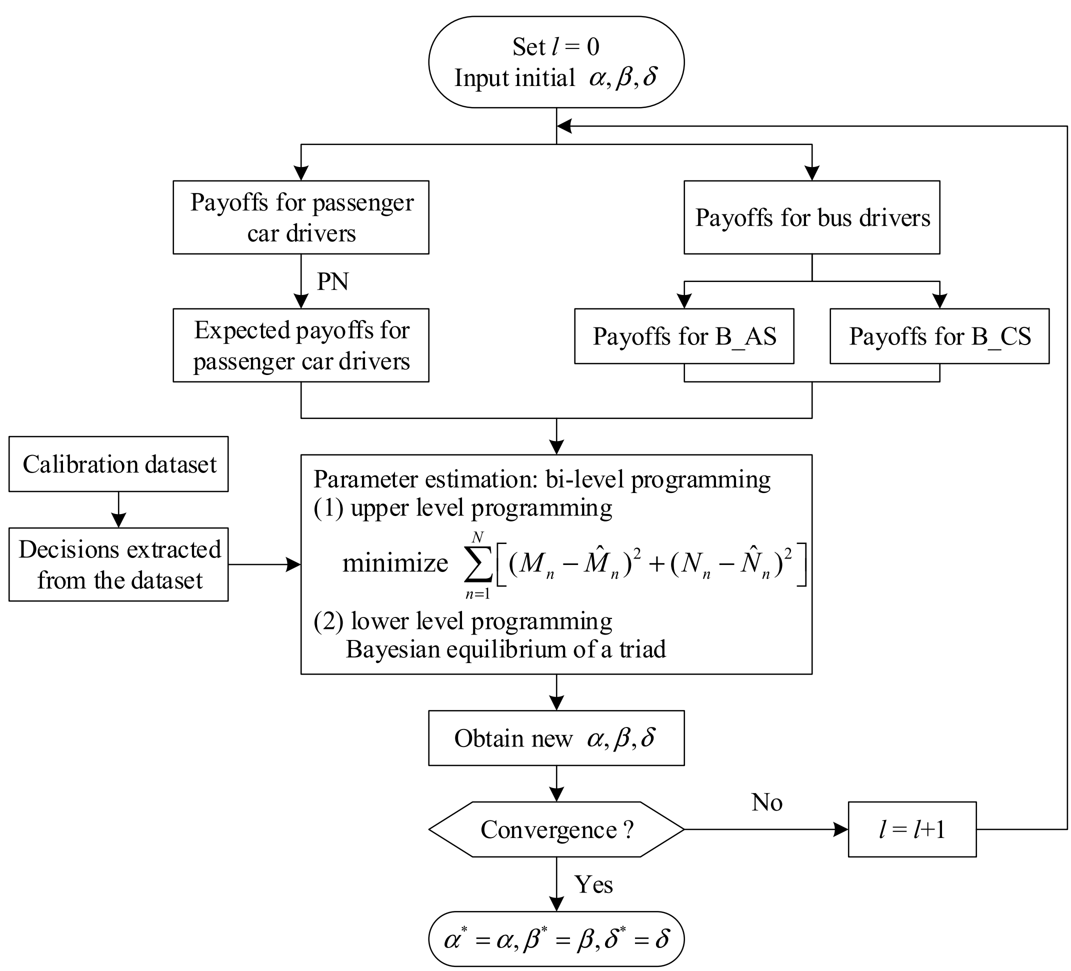


Fig. 3. Framework of parameter calibration.

be calculated. More specifically, the predicted accuracy rates of giving way and not giving way for the passenger car driver, and the predicted accuracy rates of changing a lane and not changing a lane for B_AS and B_CS are included.

In addition, the performance of the formulated model can also be evaluated by the time error of a lane changing behavior. The time error is the time difference between the observed and predicted lane changing behaviors.

5. Case study

5.1. Data collection and processing

Data used in this research was obtained through field investigation. The selection of the location of data collection needs to follow two principles: (1) a bus bay stop without dedicated bus lanes nearby; (2) an appropriate spot, which can offer a comprehensive vision for the observation area and good conditions for the long-term video recording. As illustrated in Fig. 4, Lvboqiao bus bay stop which locates on the Northwest Road (an arterial road) in Dalian city of China was chosen for data collection. The Northwest Road is in the outbound direction, which means that the traffic flow on this road is relatively dense during evening peak hours. There are totally four lanes on the segment. Lanes 3 and 4 are both left-turn lanes. The vehicles on these two lanes will not affect and be affected by the buses that are about to leave the bus bay stop. Lanes 1 and 2 are both through lanes. The vehicles on Lane 1 and the bus undoubtedly have interactions during the process of bus lane changing, when the passenger car changes a lane from Lane 1 to Lane 2, it is another game between different passenger cars, which is not focused on in this paper. Based on the field investigation, besides go straight, a small number of vehicles on Lane 2 may change lanes to Lane 3. However, such interactions do not have impacts on the mandatory lane changing for bus exiting. Therefore, the investigation area includes the whole bus bay stop and the section of Lane 1 from 40 m ahead to 40 m afterwards the bus bay stop. The data collection was conducted during evening peak hours (from 4:30 pm to 6:30 pm), on five weekdays, in good weather, in May of 2020. The overall traffic on the Northwest Road was dense but not congested, which was suitable for data collection.

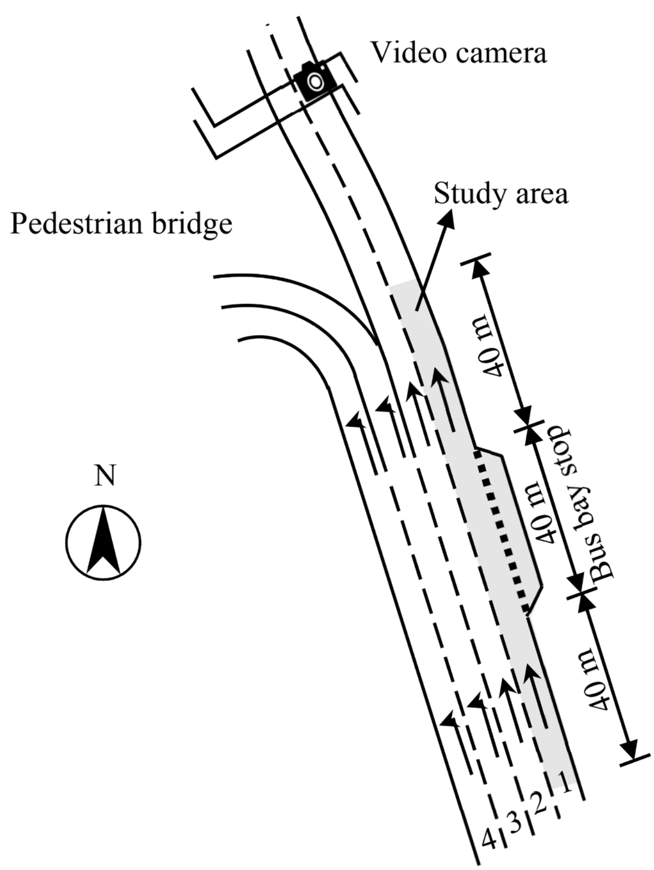


Fig. 4. Schematic illustration for the data collection site.

Data processing was carried out by applying George 2.1, a professional video processor that has been validated by Suzuki and Nakamura (2006). The accelerations, speeds and consumed time of passenger cars and buses can be collected or calculated using this processor. It is supposed that the number of passengers carried by the bus reaches the approved busload since the data were investigated during evening peak hours. Finally, 472 sets of data were obtained after video processing and data screening. According to the time sequence of the data collection, 70 % of the data will be used for the model calibration, and the remaining 30 % of the data will be applied for the model validation.

5.2. Results of calibration and validation

In the calibration, the driving style of the bus driver is selected randomly. When different PNs are given, different calibration results of the model parameters will be obtained on the basis of different payoffs for the players. The results of parameter calibration with PN from 0.1 to 0.9 are listed in Table 3.

According to the calibrated parameters, the payoffs for the passenger car and bus drivers for different strategy profiles can be calculated when PN varies from 0.1 to 0.9. Then, the behaviors for the passenger car and bus drivers can be predicted. On average, the MAE is 0.27 and 0.16 for the passenger car and bus drivers, respectively. The accuracy rates for the players' behaviors and their strategies are listed in Table 4.

From Table 4, it can be found that the formulated model is capable of predicting the behaviors of the passenger car and bus drivers. The accuracy rates for different PNs for the passenger car driver are over 70 %, and those for the bus driver are more than 80 %. Furthermore, the accuracy rates for the bus driver are classified and analyzed by considering driving styles. The accuracy rates for different PNs and driving styles for the bus driver are all more than 80 %. The accuracy rates for different strategies for the passenger car and bus drivers are more than 80 % except not giving way and other minor situations. From the perspective of providing guidance for drivers, the conservative strategy (i.e., the strategy of giving way) is helpful to guarantee the safety of passenger car drivers and reduce the total person delay.

To further test the performance of the formulated model, the time errors of lane changing for B_AS and B_CS are calculated. The predicted lane changing time depends on the prediction results of behavioral decision-makings together with T_B and T_S. The mean time error for all the bus drivers is 0.76 s, and those for B_AS and B_CS are 0.73 s and 0.79 s, respectively. The paired *t*-test is utilized to examine the significance of the time error between the observed and predicted behaviors. At a 95 % confidence level, the time error is statistically significant.

In our previous research, the two-player, non-zero-sum, non-cooperative mixed strategy game is applied to examine the lane changing issue for bus exiting, but the driving styles are not considered (Yao et al, 2021). The results in these two models are listed in Table 5. From Table 5, it can be found that the maximum accuracy rates in the incomplete information game are higher than those in the mixed strategy game except that of the strategy of not giving way. Compared with the mixed strategy game, the average accuracy rates decrease slightly in the incomplete information game, but they are already satisfactory. The lowest average accuracy rate of the strategy of not giving way results from maximizing the safety of passenger cars in the payoffs. The prediction of the strategy of giving way is advantageous to form Pareto optimal solution of giving way and changing a lane.

We have found that passenger car drivers are more inclined to the strategy of not giving way, and bus drivers (no matter what driving style they have) show more tendencies to the strategy of not changing lanes.

Table 3
Results of parameter calibration with PN from 0.1 to 0.9.

Player	Payoff symbol	Parameter	PN								
			0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
S	R_{11}	$\alpha_{11,0}$	−19.25	−13.93	−2.75	−21.11	4.47	0.50	3.95	10.82	−8.64
		$\alpha_{11,1}$	−21.67	−11.40	4.39	3.87	17.12	17.83	15.73	−4.40	8.64
	R_{12}	$\alpha_{12,0}$	−3.26	−8.34	2.97	17.98	−19.73	−4.13	−26.40	−2.64	−19.55
		$\alpha_{12,1}$	−5.57	−0.69	7.58	19.17	−4.11	13.50	−5.00	−16.91	5.67
	R_{21}	$\alpha_{21,0}$	1.37	35.01	11.76	−2.92	10.22	−3.63	16.93	−15.22	−2.80
		$\alpha_{21,1}$	1.69	−4.98	1.43	−3.87	15.24	−11.61	−2.42	5.79	16.81
		$\alpha_{21,2}$	21.10	13.23	6.12	8.21	−1.81	0.15	9.35	−4.53	2.57
	R_{22}	$\alpha_{22,0}$	2.45	23.81	2.27	19.02	−1.50	9.91	11.23	6.97	7.05
		$\alpha_{22,1}$	7.34	2.59	8.28	−4.77	−5.70	−0.87	4.10	4.08	23.18
B_AS	P_{11}	$\beta_{11,0}$	1.06	−2.95	0.74	−13.71	20.48	10.54	19.60	23.91	3.63
		$\beta_{11,1}$	−6.61	−6.64	−3.84	−5.27	5.79	1.38	3.13	0.39	14.85
	P_{12}	$\beta_{12,0}$	21.59	24.58	8.59	8.43	−6.08	9.12	3.02	−11.89	−1.08
		$\beta_{12,1}$	−4.41	3.85	−1.82	2.49	−9.18	−2.07	−4.73	−9.37	2.31
	P_{21}	$\beta_{21,0}$	−3.10	−7.48	14.66	17.11	−28.53	−2.21	−14.48	−29.37	−9.73
		$\beta_{21,1}$	14.76	7.15	10.20	1.20	−3.88	−11.99	−5.31	−8.43	−15.05
		$\beta_{21,2}$	5.25	10.78	−4.28	0.31	−7.37	−7.67	−2.16	−0.89	13.30
	P_{22}	$\beta_{22,0}$	−20.14	5.74	−22.93	−20.70	−6.81	7.29	3.68	2.68	5.41
		$\beta_{22,1}$	20.15	2.13	−7.32	−13.19	−11.84	0.49	1.35	−1.11	11.98
B_CS	Q_{11}	$\delta_{11,0}$	15.02	2.63	−6.90	−2.21	6.92	2.95	1.91	5.14	12.32
		$\delta_{11,1}$	16.20	−1.94	6.43	3.47	−11.81	5.60	−11.08	3.20	−12.09
	Q_{12}	$\delta_{12,0}$	−2.41	−7.82	9.36	15.25	−9.67	−11.02	−9.39	7.94	−2.18
		$\delta_{12,1}$	−7.83	−14.30	−2.21	−20.64	17.16	0.59	21.99	0.71	5.63
	Q_{21}	$\delta_{21,0}$	−3.90	9.21	14.60	−11.05	−11.12	5.04	−4.90	3.48	7.44
		$\delta_{21,1}$	23.08	8.69	−5.98	16.43	10.48	1.58	−4.26	−2.92	−17.78
		$\delta_{21,2}$	10.07	9.45	−10.27	12.74	−3.42	8.79	−13.05	9.36	−0.69
	Q_{22}	$\delta_{22,0}$	11.77	−8.62	9.50	2.77	−6.13	8.82	11.09	10.33	20.33
		$\delta_{22,1}$	20.84	2.09	15.55	2.41	−8.03	−7.31	−6.40	−3.00	−4.70

Table 4
Accuracy rates for the players' behaviors and their strategies.

Accuracy rates (%)		PN									
		0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	Average
Player	S	75.35	73.24	74.65	71.83	71.13	72.54	74.65	73.24	70.42	73.01
	B	85.92	85.21	84.51	85.21	80.99	84.51	84.51	80.99	87.32	84.35
	B_AS	100.00	89.29	83.72	85.96	81.69	84.71	85.86	80.70	87.50	86.60
	B_CS	84.38	84.21	84.85	84.71	80.28	84.21	81.40	82.14	85.71	83.54
Strategy	S_1	93.75	96.88	87.50	90.63	93.75	90.63	90.63	93.75	93.75	92.36
	S_2	70.00	66.36	70.91	66.36	64.55	67.27	70.00	67.27	63.64	67.37
	B_1 (B_AS)	100.00	50.00	78.26	90.00	78.57	82.14	83.33	79.07	92.00	81.49
	B_2 (B_AS)	100.00	100.00	90.00	83.78	83.72	85.96	86.96	81.69	84.62	88.53
	B_1 (B_CS)	85.42	89.36	93.33	81.82	84.00	84.00	65.22	70.00	100.00	83.68
	B_2 (B_CS)	83.75	80.60	81.16	86.54	78.26	84.38	100.00	88.89	81.82	85.04

Table 5
Contrast of results from different game models.

Accuracy rates (%)		Mixed strategy game	Incomplete information game		
			Maximum	Minimum	Average
Strategy	S_1	93.75	96.88	87.50	92.36
	S_2	80.91	70.91	63.64	67.37
	B_1	92.45	100.00	50.00	82.59
	B_2	95.51	100.00	78.26	86.79

However, the characteristics of drivers' behavioral decision-makings with different PNs are not explored in the previous research. Actually, the quantities of the Bayesian equilibria of the four strategy profiles present different characteristics with different PNs, as illustrated in Fig. 5.

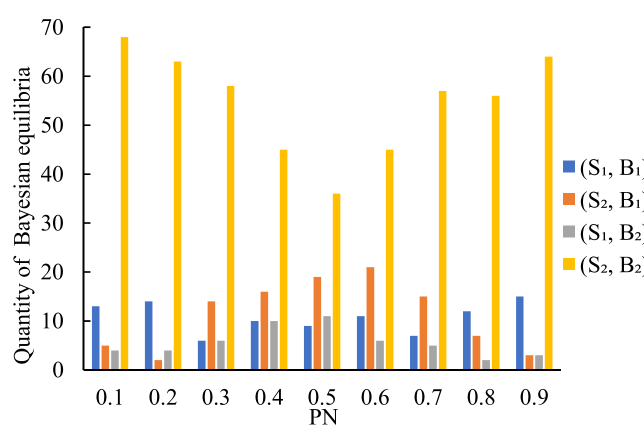


Fig. 5. Quantities of Bayesian equilibria with different PNs.

The blue bar presents the quantity of the strategy profile of giving way and changing a lane in the Bayesian equilibria and shows a steady trend in general. This strategy profile is efficient for both the passenger car and bus drivers from the perspective of the overall payoff. Even if the passenger car driver will lose some time, the total person delay decreases because of the mass passengers carried by the bus. Due to the passenger car and bus drivers' tendencies of choosing the strategies, this strategy profile is rarely selected.

The orange bar represents the quantity of the strategy profile of not giving way and changing a lane in the Bayesian equilibria. The quantity of this strategy profile shows a trend of growth first and then decline. When PN is relatively small (e.g., 0.1 and 0.2) or large (e.g., 0.8 and 0.9), the quantity of this strategy profile is limited. When the probabilities of B_AS and B_CS are close or even equal, the quantity of this strategy profile is relatively large.

The grey bar refers to the quantity of the strategy profile of giving way and not changing a lane in the Bayesian equilibria. The changing trend of the grey bars is similar to that of the orange bars, but is more stable. It can be revealed that the quantity of this strategy profile is small under every PN. For the passenger car and bus drivers, this strategy profile is helpful to guarantee traffic safety during the process of bus lane changing. However, from the perspective of traffic efficiency, this strategy profile is inefficient for rational players.

The yellow bars imply that the quantity of the strategy profile of not giving way and not changing a lane drops first and then rises. The quantity of this strategy profile is obviously larger than those of other three strategy profiles under every PN. The passenger car and bus drivers' tendencies of choosing the strategies are further verified. It can be found that a considerable number of B_AS tend to choose the strategy of not changing a lane, although the number of such drivers is not certainly as large as that of B_CS.

5.3. Relationship between PN and accuracy rate

Generally, as PN grows, the quantities of the strategies of changing a lane and not changing a lane should increase for B_AS, and decrease for B_CS. The quantities of the strategies for B_AS and B_CS are highly related since the total quantities are fixed. Thus, only the quantity of the strategies for B_AS is analyzed here. The quantities and growth rates of the strategies for B_AS with different PNs are illustrated in Fig. 6.

It can be seen from Fig. 6(a) that the quantities of the strategies of changing a lane and not changing a lane show upward trends overall. However, there is a significant increase in the quantity of the strategy of changing a lane and a decrease in the quantity of the strategy of not changing a lane when PN is between 0.2 and 0.3. Accordingly, there are a sharp rise and a sharp fall for the growth rates of the strategies of changing a lane and not changing a lane, respectively when PN is between 0.2 and 0.3, as illustrated in Fig. 6(b).

The variations of the quantities of the strategies discussed above may be related to the actual PN in the dataset. On the basis of practical experience and common sense, B_AS may take actions such as sudden acceleration and rapid insertion into traffic to pursue travel efficiency. In this case, it may cause non-collision injuries to bus passengers and the start acceleration of the bus is usually greater than 2 m/s^2 (Karekla and Tyler, 2018). If the bus drivers' driving styles are classified based on whether the start acceleration of the bus is greater than 2 m/s^2 or not, B_AS accounts for 21 % in both the calibration and validation datasets. When taking PN as 0.21, the results of parameter calibration and the accuracy rates for the players' strategies are presented in Tables 6 and 7, respectively.

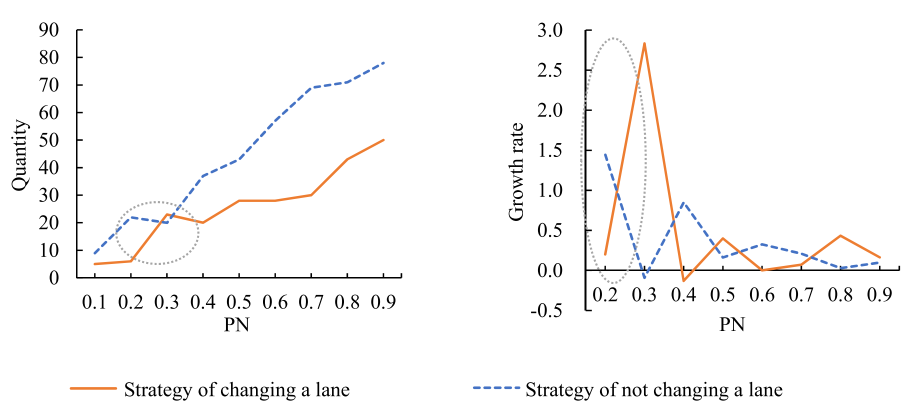
The accuracy rates for different strategies listed in Table 7 are much higher than or very close to the average accuracy rates generated by different PNs shown in Table 4. It can be inferred that the closer the predetermined PN is to the actuality, the better the model will perform. Actually, the increasing quantity of the strategy of changing a lane from 0.20 to 0.21 accounts for 41.2 % of the total growth from 0.20 to 0.30. When PN equals to 0.21, compared with 0.2 and 0.3, the quantity of the strategy of not changing a lane is the lowest.

6. Discussions

In this paper, the lane changing behaviors for bus exiting at bus bay stops are investigated using the incomplete information game by vehicle trajectory data. In the case study, the performance of the model is validated by MAE, accuracy rates for different strategies, and paired *t*-test. The characteristics of behavioral decision-makings with different PNs, and the relationship between PN and accuracy rate are analyzed deeply. The findings can provide assistance for drivers to make better decisions.

Compared with previous research, in which scholars commonly consider lane changers as the only unit of analysis using the rule-based model (Gipps, 1986; Yang and Koutsopoulos, 1996; Hidas, 2002, 2005; Kesting et al., 2007) and the utility theory-based model (Sun and Elefteriadou, 2014; Zhou et al., 2019; Pang et al., 2020), this research employs the game theory to fully explore the interactions between different vehicles. Furthermore, existing literatures focus on the lane changing game in the context of highway (Kita, 1999; Kang and Rakha, 2017; Arbis and Dixit, 2019; Talebpour et al, 2015; Ali et al, 2019). However, the traffic conditions on urban roads are more complex than those on highways, which brings difficulties to the decision-makings in the lane changing. Some scholars analyze the lane changing game in the context of urban transportation, e.g., the electrical bicycles lane changing game (Shao et al, 2018) and the lane changing game for bus entering (Ba and Liu, 2019), but there is still a lack of research on the mandatory lane changing game for bus exiting. This study fills the gap of lane changing game in urban traffic.

More specifically, based on our previous research on the lane changing issue for bus exiting at bus bay stops (Yao et al, 2021), besides the tendencies of choosing strategies, the characteristics of choosing strategies under different PNs are further examined in this paper. When



(a) Quantity (b) Growth rate

Fig. 6. Quantities and growth rates of strategies for B_AS with different PNs.

Table 6
Results of parameter calibration when taking PN as 0.21.

Parameter symbol	Parameter value for S	Parameter symbol	Parameter value for B_AS	Parameter symbol	Parameter value for B_CS
$\alpha_{11,0}$	1.11	$\beta_{11,0}$	−4.04	$\delta_{11,0}$	−13.76
$\alpha_{11,1}$	−6.49	$\beta_{11,1}$	−3.35	$\delta_{11,1}$	5.20
$\alpha_{12,0}$	12.50	$\beta_{12,0}$	10.60	$\delta_{12,0}$	−6.16
$\alpha_{12,1}$	14.00	$\beta_{12,1}$	6.80	$\delta_{12,1}$	−1.03
$\alpha_{21,0}$	−11.85	$\beta_{21,0}$	13.89	$\delta_{21,0}$	1.26
$\alpha_{21,1}$	−2.62	$\beta_{21,1}$	16.63	$\delta_{21,1}$	3.45
$\alpha_{21,2}$	−3.24	$\beta_{21,2}$	15.42	$\delta_{21,2}$	2.49
$\alpha_{22,0}$	−6.77	$\beta_{22,0}$	−10.88	$\delta_{22,0}$	−3.67
$\alpha_{22,1}$	−6.85	$\beta_{22,1}$	2.84	$\delta_{22,1}$	3.82

Table 7
Accuracy rates for passenger car and bus drivers' strategies when taking PN as 0.21.

Statistical results	S		B_AS		B_CS	
	S_1	S_2	B_1	B_2	B_1	B_2
Predicted value	30	73	11	16	37	60
Observed value	32	110	13	17	40	72
Accuracy (%)	93.75	66.36	84.62	94.12	92.50	83.33
Overall accuracy (%)	72.54		90.00		86.61	

the quantity of B_AS is relatively close to that of B_CS, the strategy of giving way and that of changing a lane are promoted for passenger car and bus drivers, respectively. Meanwhile, under some PNs, such as when aggressive or conservative bus drivers are in the majority, the prediction accuracy increases, and under other PNs, in already satisfactory cases, there is a small drop in accuracy. Through analyzing the relationship between PN and accuracy rate, it can be revealed that the more accurate the value of PN is, the better the performance of the model is. Compared with our previous study (Yao et al, 2021), the research on this issue is enriched.

7. Conclusions

Considering the existing research gaps, this study aims at exploring the mandatory lane changing behaviors for bus exiting at bus bay stops by classifying bus drivers' driving styles into aggressive and conservative driving styles. A two-player, non-zero-sum, non-cooperative game model with incomplete information is developed to describe the game between the passenger car and bus drivers. The formulated model is calibrated using the bi-level programming approach, by which the Bayesian equilibrium, as the lower-level programming, is applied to minimize the error between the observed and predicted behaviors.

The main findings of the research include:

- The proposed model can forecast the behavioral decision-makings for passenger car and bus drivers with high accuracy, which can be revealed by the results of MAE, accuracy rates for different strategies, and paired *t*-test.
- The passenger car drivers are inclined to aggressive strategies, i.e., the strategy of not giving way. The bus drivers show conservative tendencies, and they are prone to the strategy of not changing a lane.
- The distribution of quantities of strategy profiles displays different characteristics with different PNs. The consideration of driving styles can deeply capture the characteristics of drivers' decision-makings.
- The quantity of the strategy profile of not giving way and not changing a lane is the majority, which further verifies the drivers' tendencies of choosing strategies. In this case, even if when PN is high, a significant number of bus drivers still choose the strategy of not changing a lane due to the aggressiveness shown by the passenger car drivers.
- When the number of B_AS and that of B_CS are close, the quantity of the strategy profile of not giving way and changing a lane increases, thus traffic conflicts are easily caused.
- The quantity of the strategy profile of giving way and changing a lane is stable in general with different PNs.
- In the four strategy profiles, the quantity of the strategy profile of giving way and not changing a lane is the least. When PN is close to 0.5, the quantity of such a strategy profile rises, and the trend is similar to that shown by the strategy profile of not giving way and changing a lane.
- In view of the relationship between safety and the start acceleration of the bus, the in-depth analysis of the passenger car and bus drivers' behaviors and the changes of the strategy quantities are carried out. The outcomes reveal that the higher the accuracy of PN is, the better the prediction results are.

7.1. *Limitations and future research scope*

The players in the formulated model are assumed to be completely rational. However, besides the driving style, the driver's background information (e.g., gender, age and driving experience) is worth analyzing in details in the mandatory lane changing model. Traffic has different characteristics in different countries, such as traffic flow status, traffic rules and traffic policies, which may lead to different characteristics of driving behavioral decision-makings. The model proposed in this paper for the lane changing behavior for bus exiting at bus bay stops is suitable for general situations. It is better to obtain relevant data from other cities and countries in future research to further investigate the characteristics of behavioral decision-makings. In addition, more game models can be built to examine discretionary lane changing of passenger cars near bus bay stops, and the interactions between buses and non-motor vehicles.

7.2. *Practical implications*

The model proposed in this paper and the related methods can provide significant theoretical references for the behavioral decision-makings of passenger car and bus drivers. The findings can also reduce the negative impacts on traffic flow operations caused by the mandatory lane changing for bus exiting, and improve traffic safety and efficiency. Furthermore, through predicting behavioral decision-makings of passenger car and bus drivers, the traffic organization around bus bay stops can be optimized. The simulation and implementation of the lane changing behaviors for connected and autonomous vehicles with emerging technologies have to rely on the human's actual behavioral decision makings. The research on the behavioral decision makings of lane changing for human-driven vehicles can help to better understand and design the vehicle driving assistance system for connected and autonomous vehicles.

CRediT authorship contribution statement

Ronghan Yao: Conceptualization, Methodology, Resources, Writing – review & editing, Visualization, Supervision, Project administration, Funding acquisition. **Xiaojing Du:** Conceptualization, Methodology, Software, Investigation, Data curation, Writing – original draft, Visualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The research reported in this article has been funded by the National Natural Science Foundation of China (Grant No. 52172314) and the Fundamental Research Funds for the Central Universities of Ministry of Education of China (Grant No. DUT20JC40).

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